

iA

2025
ianie

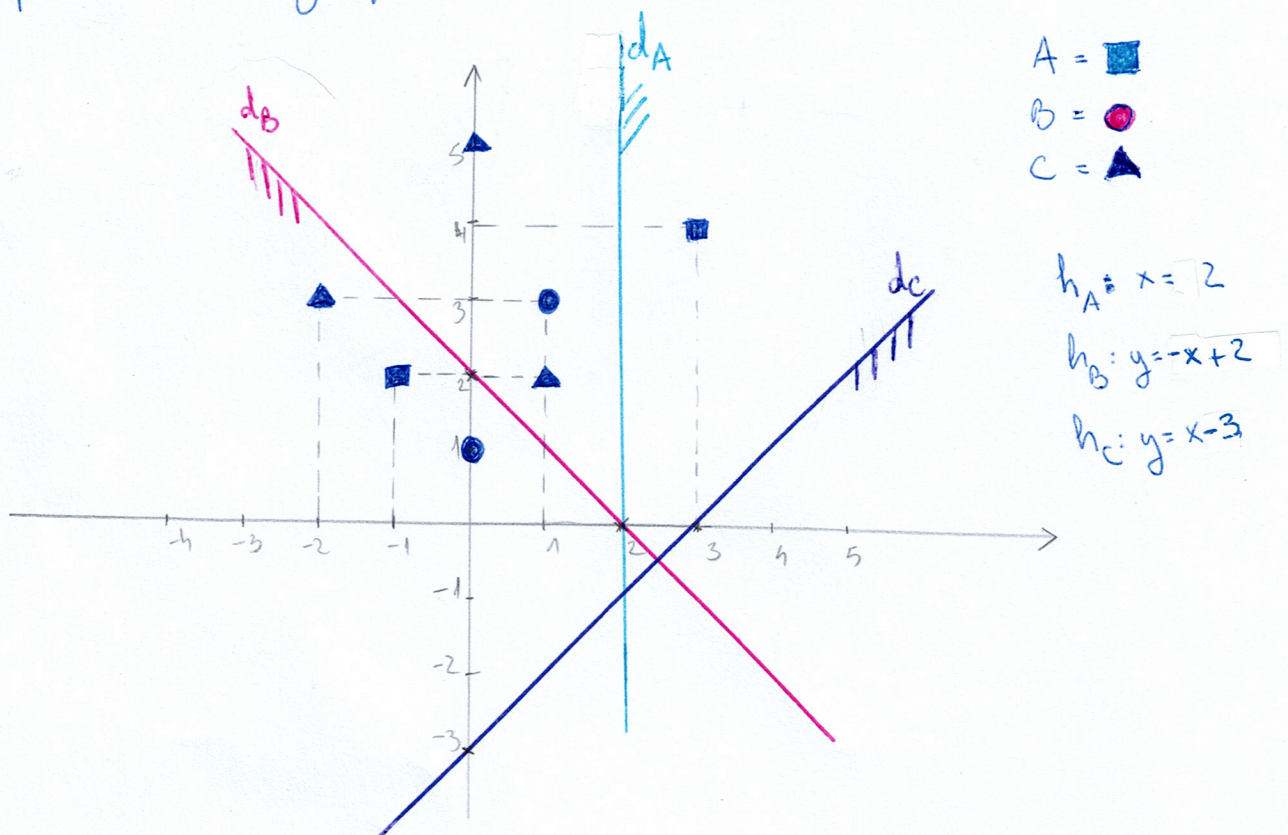
Test seminar v2

$$1. S = \{ ([-1, 2], A), ([3, 4], A), ([0, 1], B), ([1, 3], B), \\ ([0, 5], C), ([1, 2], C), ([-2, 3], C) \}$$

Se dau hiperplanele:

$$w_A = [1, 0] \quad b_A = -2$$
$$w_B = [-1, -1] \quad b_B = 2$$
$$w_C = [1, -1] \quad b_C = -3$$

a) Reprezentare grafică



b) Funcția de pierdere Hinge

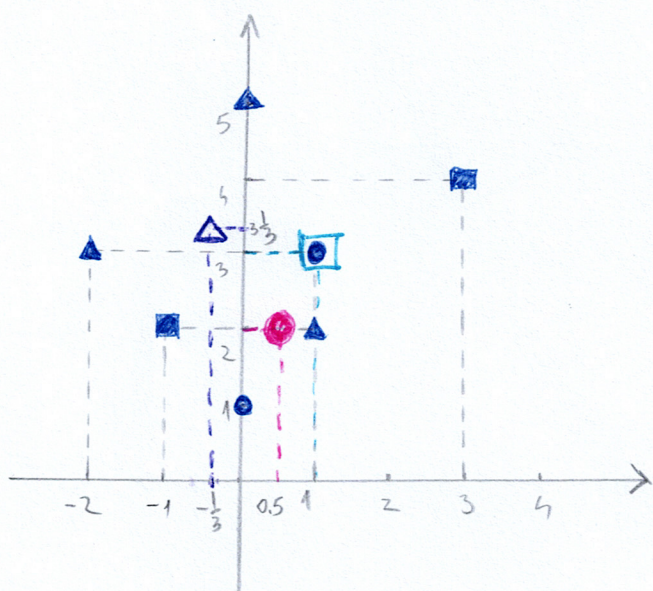
$$L_i = \sum_{j \neq z_i} \max(0, s_j - s_{z_i} + 1)$$

$z \neq z$

x	y	z	□	○	△	$\hat{z}$	"scor"	Loss	Formula
-1	2	■	-3	1	-6	○	1	5	$1 - (-3) + 1$
3	4	■	1	5	-4	○	5	5	$5 - 1 + 1$
0	1	●	-2	1	-4	○	1	0	0
1	3	●	-1	-2	-5	□	-1	2	$-1 - (-2) + 1$
0	5	▲	-2	-3	-8	□	-2	13	$-2 - (-8) + 1 - 3 - (-8) + 1$
1	2	▲	-1	-1	-4	□/○	-1	8	$-1 - (-4) + 1 - 1 - (-4) + 1$
-2	3	▲	-4	1	-8	○	1	15	$-4 - (-8) + 1 + 1 - (-8) + 1$

$48/7 \leftarrow$ Hinge Loss (avg.)

c) Calculați centrulidul fiecărei clase și aplicați 1-NN bazat pe distanța L_1 .



$$x_A = \frac{-1+3}{2} = 1 \quad y_A = \frac{2+4}{2} = 3$$

$$x_B = \frac{0+1}{2} = 0.5 \quad y_B = \frac{1+3}{2} = 2$$

$$x_C = \frac{-2+0+1}{3} = -\frac{1}{3} \quad y_C = \frac{2+3+5}{3} = 3\frac{1}{3}$$

1-NN predictions

x	y	z	$\hat{z}$	Acc.
-1	2	■	△	x
3	4	■	□	✓
0	1	●	○	✓
1	3	●	□	x
0	5	▲	△	✓
1	2	▲	○	x
-2	3	▲	△	✓

4/8 corecte
= 50% acuratețe

$$d((-1, 2), \blacksquare) = 1 - 1 - 1 + 1 + 2 - 3 = 3$$

$$d((-1, 2), \blacktriangle) = 1 - 1 + \frac{1}{3} + 1 + 2 - \frac{10}{3} = 2$$

d) Eroarea de clasificare este de 50%.

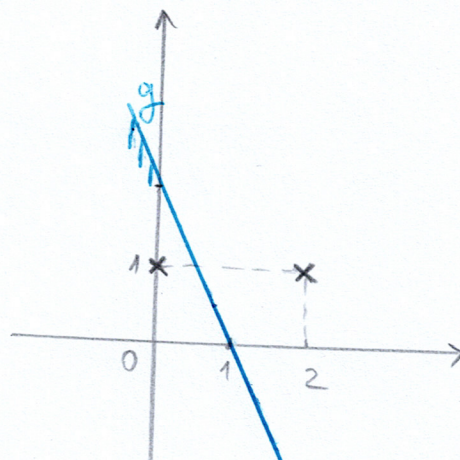
$$2. S = \{ ([0,1], 0), ([2,1], 0) \}$$

$$g(x) = w \cdot x + b$$

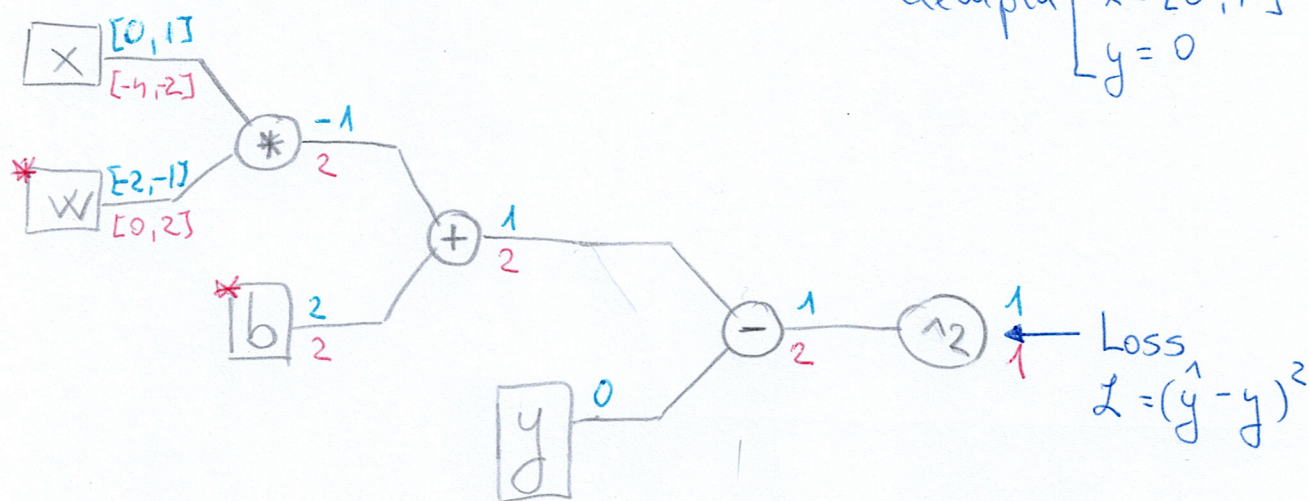
$$w = [-2, -1]$$

$$b = 2$$

$$d_g: y = -2x + 2$$



a) Graful computațional + b) Valorile gradientilor pentru primul exemplu $\begin{cases} x = [0, 1] \\ y = 0 \end{cases}$



c) Aplicăm SGD pentru primul exemplu cu $\eta = 0.2$.

$$w \leftarrow w - \eta \cdot \frac{\partial L}{\partial w} = [-2, -1] - 0.2 \cdot [0, 2]$$

$$w \leftarrow [-2, -1.4]$$

$$b \leftarrow b - \eta \cdot \frac{\partial L}{\partial b} = 2 - 0.2 \cdot 2$$

$$b \leftarrow 1.6$$

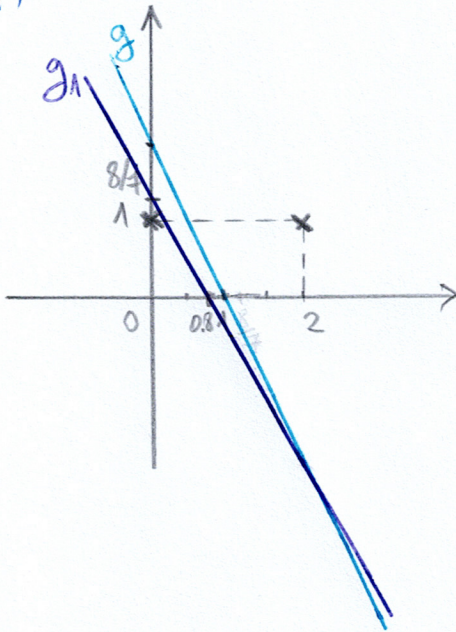
$$\frac{\partial L}{\partial \hat{y}} = 2(\hat{y} - y)$$

$$\frac{\partial L}{\partial b} = \frac{\partial \hat{y}}{\partial b} \frac{\partial L}{\partial \hat{y}} = 2(\hat{y} - y)$$

$$\frac{\partial L}{\partial w} = \frac{\partial \hat{y}}{\partial w} \frac{\partial L}{\partial \hat{y}} = x \cdot 2(\hat{y} - y)$$

$$\frac{\partial L}{\partial x} = \frac{\partial \hat{y}}{\partial x} \frac{\partial L}{\partial \hat{y}} = w \cdot 2(\hat{y} - y)$$

d)



$$W_1 = [-2, -1.4]$$

$$b_1 = 1.6$$

$$g_1: -2x - 1.4y + 1.6 = 0$$

$$g_1: 1.4y = -2x + 1.6$$